Drug–Drug Interaction Prediction

Technical Report for the KG-SS-GNN Project

Course

CSE 433 – Emerging Topics in Computer Science

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Abstract

This report documents the deep learning project implemented in `ddi-phase2_v2.ipynb` and modularized in `project/repo`. The project predicts drug–drug interaction types from pairs of molecular SMILES strings using the local TDC DrugBank dataset downloaded to `project/data`. The final model is KG-SS-GNN: a Knowledge-Graph-enhanced Substructure-aware Siamese Graph Neural Network. It combines RDKit molecular graphs, BRICS/Morgan substructure fingerprints, 141-dimensional knowledge-graph proxy features, a shared Graph Attention Network encoder, cross-drug co-attention, and an MLP classifier. The notebook compares KG-SS-GNN against two baselines, Siamese GCN and DeepDDI, on a cold-drug split. KG-SS-GNN achieves the best reported test performance: Accuracy 0.4900, Macro-F1 0.3598, Recall 0.4184, AUROC 0.9189, and AUPRC 0.5130.

1 Problem Definition

Drug–drug interactions (DDIs) occur when one drug changes the effect, concentration, toxicity, or safety profile of another drug. This is clinically important because patients often take multiple drugs at the same time, especially in chronic disease management. A harmful DDI may increase toxicity, reduce treatment effectiveness, increase bleeding risk, cause cardiac side effects, or create central nervous system effects.

The computational task in this project is multiclass DDI prediction. Given two drugs D_1 and D_2, each represented by a SMILES string, the model predicts the interaction class y:

$$f(D_1, D_2) \to y, \qquad y \in \{1, \dots, 86\}.$$

The input is not a molecule image. Each SMILES string is parsed by RDKit into a chemical graph where atoms are nodes and bonds are edges. The output is one of the DrugBank interaction types. This makes the project a supervised multiclass classification problem under strong class imbalance.

1.1 Why This Problem Needs Deep Learning

The number of possible drug pairs grows quadratically with the number of drugs. Testing all combinations experimentally is expensive and slow. Rule-based systems can miss subtle molecular and biological patterns. Deep learning is useful here because it can combine several sources of information:

- atom-level molecular graph structure;

- chemically meaningful substructures;
- drug-pair interaction patterns;
- pharmacokinetic and biological proxy features;
- nonlinear feature interactions learned from data.

1.2 Main Research Question

The main research question is:

> Can a substructure-aware Siamese GNN enhanced with KG-proxy features and cross-drug co-attention outperform graph-only and fingerprint-only baselines on cold-drug DDI prediction?

The notebook answers this experimentally by training three models on the same split and reporting the same metrics for all of them.

2 Literature Review

2.1 DeepDDI

Ryu, Kim, and Lee proposed DeepDDI, a deep learning framework for predicting drug–drug and drug–food interactions [1]. DeepDDI represents drug pairs with structural information and predicts one of the DrugBank interaction types. The important contribution is that it treats DDI prediction as multiclass interaction-type prediction rather than binary interaction detection.

This project uses a DeepDDI-style baseline. Each drug is represented by a fixed-length fingerprint, the two fingerprints are concatenated, and an MLP predicts the interaction type. This baseline is fast and simple, but it loses fine atom-level structural information.

2.2 Decagon

Zitnik, Agrawal, and Leskovec introduced Decagon, a graph convolutional model for polypharmacy side-effect prediction [2]. Decagon builds a multimodal biomedical graph containing drugs, proteins, drug–protein links, protein–protein links, and side-effect relations. It performs multi-relational link prediction over many side-effect types.

Decagon motivates the knowledge-enhanced part of this project. Many DDIs are not caused only by molecular shape. They may happen because drugs share enzymes, targets, transporters, or metabolic pathways. The project therefore adds KG-proxy features:

Morgan fingerprints at different radii plus normalized pharmacokinetic and physicochemical descriptors.

2.3 SSI-DDI

Nyamabo, Yu, and Shi proposed SSI-DDI, which predicts DDIs through substructure–substructure interactions [3]. Its key idea is that interactions are often caused by local molecular fragments rather than whole molecules. SSI-DDI uses co-attention to model which substructures of one drug interact with substructures of another drug.

This motivates two components in KG-SS-GNN: BRICS/Morgan substructure fingerprints and cross-drug co-attention. The final model does not merely encode each drug independently; it explicitly computes a pairwise co-attended representation.

2.4 SA-DDI and Substructure-Aware GNNs

Yang et al. proposed SA-DDI, a substructure-aware graph neural network for explainable DDI prediction [4]. The paper emphasizes that combining molecular graph learning with substructure-level reasoning improves both prediction and explanation. This project follows the same direction by fusing GAT-based graph embeddings with BRICS/Morgan fingerprints before pairwise attention.

2.5 Research Gap Addressed

The reviewed literature suggests three requirements for a strong DDI model:

1. use molecular structure rather than drug IDs only;
2. model pairwise interaction between the two drugs;
3. include biological or pharmacokinetic context beyond raw structure.

The KG-SS-GNN implementation addresses these requirements using a shared GAT encoder, BRICS/Morgan substructure features, KG-proxy descriptors, and cross-drug co-attention.

3 Dataset Description

3.1 Local Dataset Files

The dataset was downloaded locally under:

`project/data`

Two files are present:

- `drugbank_ddi.csv`: TDC-style CSV used by the repository;
- `drugbank.tab`: tab-separated TDC source file with interaction templates.

The CSV contains 191,808 rows and five columns, summarized in Table 1.

Table 1: Dataset columns and descriptions.

Column	Meaning
`Drug1_ID`	DrugBank ID for the first drug.
`Drug1`	SMILES string for the first drug.
`Drug2_ID`	DrugBank ID for the second drug.
`Drug2`	SMILES string for the second drug.
`Y`	Integer interaction class label.

The local CSV has 86 unique interaction labels and 1,706 unique drugs. The label distribution is highly imbalanced. The most frequent label appears 60,751 times, while the rarest label appears only 6 times. This imbalance is a central reason for using focal loss and weighted sampling.

3.2 Notebook Split

The notebook uses the TDC DrugBank cold-drug split. The notebook outputs report the split shown in Table 2.

Table 2: Cold-drug split used in the notebook.

Split	Pairs
Training	94,892
Validation	1,835
Test	7,527
Total used in split	104,254

The cold-drug split is stricter than a random split. Some drugs are withheld from training, so the model is evaluated on interactions involving unseen drugs. This is more realistic for drug discovery than a random pair split.

3.3 Class Imbalance

DrugBank DDI labels are not uniformly distributed. Common interaction types dominate the training set, while clinically important rare interactions may have few examples. If plain cross-entropy is used, the model can obtain reasonable accuracy by focusing on common classes while performing poorly on rare classes. This is why Macro-F1, macro recall, focal loss, and weighted sampling are important in this project.

4 Feature Engineering

4.1 Molecular Graph Features

Each SMILES string is parsed by RDKit into a graph. Atoms are nodes and bonds are edges. Each atom is represented by a 79-dimensional vector containing:

- atomic number;
- atom degree;
- total valence;
- formal charge;
- hybridization;
- chirality;
- total attached hydrogens;
- aromaticity;
- ring membership;
- ring sizes;
- scaled atom mass;
- implicit-hydrogen flag.

Each bond is represented by a 10-dimensional vector containing bond type, conjugation, ring membership, and stereochemistry. Bonds are added in both directions so that graph message passing can move both ways along a chemical bond.

4.2 BRICS/Morgan Fingerprints

BRICS fragmentation decomposes molecules into chemically meaningful fragments. For each fragment, a Morgan fingerprint is computed. The final 512-dimensional fingerprint is obtained by max-pooling fragment fingerprints. For very large molecules, the notebook computes a whole-molecule Morgan fingerprint directly to avoid expensive BRICS processing.

4.3 KG-Proxy Features

The model introduces a 141-dimensional KG-proxy vector. It contains:

- 64-bit radius-1 Morgan fingerprint as a local environment proxy;
- 64-bit radius-3 Morgan fingerprint as an extended pharmacophore proxy;
- 13 normalized descriptors: molecular weight, LogP, TPSA, hydrogen bond donors, hydrogen bond acceptors, rotatable bonds, aromatic rings, total rings, heterocycles, fraction CSP3, radical electrons, atom count, and bond count.

These features are not a full curated biomedical knowledge graph, but they add biological and pharmacokinetic context beyond the graph encoder. The report therefore refers to them as KG-proxy features.

5 Model Architecture Diagram

The architecture is KG-SS-GNN: Knowledge-Graph-enhanced Substructure-aware Siamese Graph Neural Network.

Drug A SMILES and Drug B SMILES

$\Downarrow$ RDKit parsing

Molecular graph	BRICS/Morgan	KG-proxy vector
79-d atom features	512-d fingerprint	64+64+13
10-d bond features	per drug	141 dimensions

$\Downarrow$

Shared 3-layer GAT encoder + BRICS fusion

$\Downarrow$

Drug A embedding and Drug B embedding

$\Downarrow$

Cross-drug co-attention

$\Downarrow$

Concatenate: $[h_A, h_B, h_{co}, k_A, k_B]$

$\Downarrow$

MLP classifier

$\Downarrow$

Softmax over 86 interaction classes

Figure 1: KG-SS-GNN architecture implemented in the notebook and repository.

6 Model Architecture Details

6.1 Shared GAT Encoder

The GAT encoder is shared between the two drugs, making the model Siamese. This means the same parameters are used to encode Drug A and Drug B. The encoder has three GAT layers:

- GAT layer 1: atom features to hidden representations with multiple heads;
- GAT layer 2: deeper multi-head graph attention;
- GAT layer 3: single-head output to a 128-dimensional drug embedding.

Layer normalization, ELU activations, and dropout are used after the first two GAT layers. Global mean pooling converts atom-level embeddings into a drug-level graph embedding.

6.2 Substructure Fusion

The 512-dimensional BRICS/Morgan fingerprint is projected into the hidden dimension and added to the pooled GAT embedding:

$$h_D = \mathrm{GAT}(G_D) + W_f f_D.$$

This combines atom-level graph structure with fragment-level chemical substructure information before cross-drug attention.

6.3 Cross-Drug Co-Attention

For drug embeddings h_A and h_B, the model computes query, key, and value projections. Drug A attends to Drug B, and Drug B attends to Drug A:

$$\alpha_A = \sigma\left(\frac{Q(h_A)K(h_B)}{\sqrt{d}}\right), \qquad \alpha_B = \sigma\left(\frac{Q(h_B)K(h_A)}{\sqrt{d}}\right).$$

The attended representations are concatenated and projected back to the hidden dimension. This gives the model an explicit pairwise interaction mechanism.

6.4 KG Feature Projection

The 141-dimensional KG-proxy vector for each drug is projected through:

$$k_D = \mathrm{Dropout}\Big(\mathrm{ReLU}(\mathrm{LayerNorm}(W_k x_D))\Big).$$

This gives each drug a biological and physicochemical context vector aligned with the GAT embedding size.

6.5 Classifier

The representation is:

$$z = [h_A; h_B; h_{co}; k_A; k_B].$$

Since each component has hidden dimension 128, the joint vector has dimension 640. It is passed to an MLP:

$$640 \to 512 \to 256 \to 86.$$

Batch normalization, ReLU, and dropout are used inside the classifier.

7 Training Setup

The notebook trains three models:

1. Siamese GCN baseline;
2. DeepDDI-style fingerprint MLP baseline;
3. KG-SS-GNN proposed/final model.

The final model training setup is summarized in Table 3.

Table 3: Final KG-SS-GNN training setup.

Setting	**Value**
Epochs	100
Batch size	64
Optimizer	Adam
Learning rate	10^{-3}
Weight decay	10^{-4}
Scheduler	CosineAnnealingLR
Minimum learning rate	10^{-5}
Gradient clipping	Norm 1.0
Loss	Focal Loss, $\gamma = 2$
Class balancing	WeightedRandomSampler
Model selection	Best validation Macro-F1

The model saved the best validation-F1 checkpoint as:

`kgssgnn_best.pt`

The project keeps two notebook-exported pickle checkpoints for inference and deployment:

```
project/checkpoint/kgssgnn_best_pickle_cold_split.pkl
project/checkpoint/kgssgnn_best_pickle_semi_cold_split.pkl
```

The `cold_split` checkpoint corresponds to the stricter evaluation setting used throughout the report, where test interactions include drugs that were withheld from training. The `semi_cold_split` checkpoint is kept as an additional deployed variant for comparison, where the split is less strict because at least part of the pair-level drug information has been seen during training. The Streamlit demo allows selecting either checkpoint so users can compare inference behavior under the two trained variants.

8 Loss Functions Used

8.1 Focal Loss

The main loss is focal loss:

$$\mathcal{L} = -(1 - p_t)^{\gamma} \log(p_t),$$

where p_t is the predicted probability assigned to the true class. The project uses $\gamma = 2$. Focal loss down-weights easy examples and gives more relative importance to hard examples. This is useful because DrugBank has many common interaction types and a long tail of rare interaction types.

8.2 Class Weights in Baselines

The baseline models use focal loss with inverse-frequency class weights:

$$w_c \propto \frac{1}{n_c}.$$

This increases the penalty for misclassifying rare classes.

8.3 Weighted Sampling in KG-SS-GNN

The final KG-SS-GNN uses focal loss without class weights because the training loader uses `WeightedRandomSampler`. Each training sample is sampled with probability inversely proportional to its class frequency. This balances the training batches while keeping the loss numerically simpler.

9 Performance Metrics

The project reports five metrics:

- **Accuracy**: overall fraction of correct predictions;
- **Macro-F1**: mean F1 score across all classes;
- **Macro Recall**: mean recall across all classes;

- **AUROC**: multiclass one-vs-rest area under the ROC curve;
- **AUPRC**: mean one-vs-rest area under the precision-recall curve.

Macro-F1 and macro recall are critical because accuracy can hide rare-class failure. In clinical DDI prediction, missing rare but harmful interactions is more serious than confusing common low-risk classes.

10 Experimental Results

The notebook reports the cold-drug test-set results shown in Table 4.

Table 4: Test-set performance on the cold-drug split.

Model	Accuracy	Macro-F1	Recall	AUROC	AUPRC
Siamese GCN	0.1562	0.1464	0.3203	0.8936	0.3702
DeepDDI	0.1736	0.1237	0.2299	0.8245	0.2631
KG-SS-GNN	**0.4900**	**0.3598**	**0.4184**	**0.9189**	**0.5130**

KG-SS-GNN performs best on every metric. The largest gains are in accuracy, Macro-F1, and AUPRC. This supports the value of combining graph attention, substructure fingerprints, co-attention, and KG-proxy features.

10.1 Validation Training Behavior

The saved validation history for KG-SS-GNN is summarized in Table 5.

Table 5: KG-SS-GNN validation history.

Epoch	Loss	Macro-F1	AUROC
1	0.4626	0.2948	0.9056
5	0.0902	0.3674	0.9110
10	0.0746	0.3326	0.9070
20	0.0637	0.3597	0.9046
40	0.0489	0.3458	0.9103
50	**0.0410**	**0.4212**	0.9117
75	0.0233	0.3936	**0.9298**
100	0.0139	0.4211	0.9194

The best validation Macro-F1 was 0.4212 at epoch 50. The notebook restores the best checkpoint before final test evaluation.

11 Bias–Variance Analysis

11.1 Baseline Bias

The Siamese GCN baseline encodes each drug independently and then concatenates the embeddings. This introduces bias because the model does not explicitly model which features of Drug A interact with which features of Drug B. It can learn useful molecular representations, but its pairwise reasoning is weak.

The DeepDDI baseline has another source of bias: fingerprints compress molecules into fixed binary vectors. This is computationally efficient, but it can lose atom-level and bond-level patterns. Its Macro-F1 of 0.1237 indicates that this representation is not expressive enough for difficult cold-drug multiclass prediction.

11.2 Variance and Overfitting

KG-SS-GNN is more expressive than the baselines, so it has higher potential variance. The validation history shows that training loss continues decreasing after epoch 50, while Macro-F1 fluctuates. This suggests some overfitting risk after the best validation point. The notebook controls this by storing the best validation checkpoint and restoring it before test evaluation.

11.3 Effect of Cold Split

The cold-drug split increases generalization difficulty. A model cannot simply memorize frequent drug IDs. It must generalize from molecular and KG-proxy features. The higher KG-SS-GNN AUROC and AUPRC indicate that the final model generalizes better than the baselines under this split.

12 Regularization Strategy

The project uses several regularization techniques:

- **Dropout**: used in the GAT encoder, fingerprint projection, KG projection, and classifier;
- **Weight decay**: Adam uses 10^{-4} weight decay;
- **Layer normalization**: used in GAT hidden states and KG projection;

- **Batch normalization**: used in classifier hidden layers;
- **Gradient clipping**: gradients are clipped to norm 1.0;
- **Cosine annealing**: learning rate decays smoothly to 10^{-5};
- **Best-checkpoint selection**: final test evaluation uses the best validation Macro-F1 checkpoint, not simply the last epoch;
- **WeightedRandomSampler**: reduces majority-class domination.

13 Error Analysis

13.1 Class Imbalance Errors

The dataset has severe imbalance. Rare classes are difficult because the model sees few examples and may confuse them with common interaction types. The notebook directly analyzes this by plotting class size against per-class recall. Low recall for small classes indicates data scarcity rather than only architecture weakness.

13.2 Baseline Error Patterns

The Siamese GCN has better AUROC than DeepDDI but weak Macro-F1, indicating that it can rank classes somewhat well but struggles to assign exact rare labels. DeepDDI has higher accuracy than GCN but worse AUROC and recall, suggesting more dependence on common fingerprint patterns.

13.3 KG-SS-GNN Error Patterns

KG-SS-GNN improves all metrics but does not solve the problem completely. Its Macro-F1 is 0.3598, meaning many classes remain difficult. The recall of 0.4184 shows that rare or ambiguous interactions are still missed. These errors likely come from:

- rare labels with very few examples;
- noisy or broad DrugBank interaction categories;
- invalid or difficult SMILES strings;
- KG-proxy features not being a full curated biological graph;
- interactions caused by clinical context not encoded in molecular structure.

14 Limitations

The final project has the following limitations:

- The KG features are proxy features rather than a full DrugBank target, enzyme, pathway, and protein-interaction graph;
- some rare classes have extremely few examples, limiting achievable recall;
- the model predicts one class per pair even though real clinical interactions may have multiple effects;
- the Streamlit demo requires a trained checkpoint; it does not ship a fake model;
- the local repository split may differ slightly from the TDC internal cold split if TDC split metadata is not reused;
- training the full model is computationally expensive. The notebook reported 5,370 seconds for KG-SS-GNN training.

15 Future Work

Future improvements should focus on:

1. building a real biomedical knowledge graph from DrugBank targets, enzymes, transporters, and pathways;
2. adding calibration so predicted probabilities are clinically more meaningful;
3. testing multi-label prediction for pairs with multiple interaction mechanisms;
4. adding interpretability visualizations for GAT and co-attention weights;
5. running ablations without KG-proxy features, without BRICS features, and without co-attention;
6. packaging the trained checkpoint with versioned metadata.

16 Code Repository and Reproducibility

The code repository is:

`project/repo`

It contains modular code:

- `data.py`: local DrugBank loading, cold split, feature caches, and datasets;
- `checkpoints.py`: checkpoint loading for repository checkpoints and the notebook pickle checkpoint;
- `features.py`: RDKit graph, BRICS/Morgan, and KG-proxy features;
- `models.py`: Siamese GCN, DeepDDI, and KG-SS-GNN;
- `train.py`: KG-SS-GNN training;
- `evaluate.py`: checkpoint evaluation;
- `predict.py`: checkpoint-based inference;
- `app.py`: Streamlit demo.

16.1 Checkpoint Format

The training script saves a dictionary containing:

- KG-SS-GNN PyTorch state-dict tensors;
- 86-class classifier weights;
- GAT, co-attention, KG-projection, and classifier parameters.

The repository loader supports this pickle state-dict format directly and infers the number of output classes from the final classifier bias.

17 Conclusion

The project is a complete DDI prediction pipeline using local DrugBank data, molecular graph featurization, substructure fingerprints, KG-proxy features, cross-drug co-attention, and multiclass classification. The final KG-SS-GNN model outperforms the Siamese GCN and DeepDDI baselines across all reported test metrics. The strongest evidence is the improvement in Macro-F1 from 0.1464 and 0.1237 for the baselines to 0.3598 for KG-SS-GNN, and the improvement in AUPRC to 0.5130. These results support the project hypothesis that combining graph attention, substructure features, KG-proxy features, and pairwise co-attention improves cold-drug DDI prediction.

References

[1] Ryu, J. Y., Kim, H. U., and Lee, S. Y. Deep learning improves prediction of drug–drug and drug–food interactions. *Proceedings of the National Academy of Sciences*, 115(18), 2018. DOI: 10.1073/pnas.1803294115.

[2] Zitnik, M., Agrawal, M., and Leskovec, J. Modeling polypharmacy side effects with graph convolutional networks. *Bioinformatics*, 34(13), 2018. DOI: 10.1093/bioinformatics/bty294.

[3] Nyamabo, A. K., Yu, H., and Shi, J. SSI-DDI: substructure–substructure interactions for drug–drug interaction prediction. *Briefings in Bioinformatics*, 22(6), 2021. DOI: 10.1093/bib/bbab133.

[4] Yang, Z., Zhong, W., Lv, Q., and Chen, C. Y.-C. Learning size-adaptive molecular substructures for explainable drug–drug interaction prediction by substructure-aware graph neural network. *Chemical Science*, 13, 8693–8703, 2022. DOI: 10.1039/D2SC02023H.

[5] Huang, K. et al. Therapeutics Data Commons: Machine learning datasets and tasks for drug discovery and development.

[6] RDKit: Open-source cheminformatics software. https://www.rdkit.org/

[7] Kipf, T. N. and Welling, M. Semi-Supervised Classification with Graph Convolutional Networks.

[8] Lin, T. Y. et al. Focal Loss for Dense Object Detection.